

Manan Sharma

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TECHNICAL SKILLS

Programming: Python, C, C++, Rust, R, TypeScript, JavaScript, Kotlin, MATLAB, SQL, Bash/Shell, PowerShell

AI / ML & Vision: PyTorch, TensorFlow, Keras, Intel OpenVINO, OpenCV, YOLOv8, DeepLabV3+, Residual U-Net (CRUNet), Scikit-learn, Model Context Protocol (MCP), Hugging Face, LLM Pipelines

Embedded & IoT: OpenWrt, Linux Kernel Customization, Device Tree Overlays (.dts), SPI / I2C / UART Drivers, LoRaWAN (SX1303 Baseband), ChirpStack, MIPS (MT7628), ARM (NanoPi Allwinner H3), Raspberry Pi, Arduino

Systems & Cloud: Linux Administration (Debian, Ubuntu, Arch), Docker, Git, CI/CD, Azure, GCP, AWS (ECS, S3, Lambda), TCP/IP Networking, iptables, Network Security, DBMS

Full Stack & Scientific: FastAPI, Flask, React.js, Three.js (WebGL, GLSL Shaders), HTML5/CSS3, Tailwind CSS, NumPy, SciPy, MNE-Python (DSP / EEG), QGIS, Figma

EDUCATION

University of Jammu

Bachelor of Technology (B.Tech) in Artificial Intelligence & Mathematics

Jammu & Kashmir, India

2024 – Present

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Systems Programming, Linear Algebra, Discrete Mathematics, Database Management Systems (DBMS), Operating Systems, Computer Networks.

Kendriya Vidyalaya No. 1, Udhampur

Higher Secondary Education (Senior Secondary)

Jammu & Kashmir, India

Graduated 2024

EXPERIENCE

Embedded & Internet of Things Engineer (Intern) – UniConverge Technologies

Jun. 2026 – Jul. 2026

Noida, Uttar Pradesh, India

On-site

- Built custom **embedded Linux firmware** images utilizing OpenWrt build environments, establishing seamless cross-compilation interoperability across **MIPS (MT7628)** and **ARM (NanoPi Allwinner H3)** gateway architectures.
- Engineered hardware integration for **Semtech SX1303 LoRa baseband concentrators** by authoring custom device tree overlays (.dts), pin multiplexing configurations, and kernel-space SPI drivers.
- Optimized **kernel-space SPI drivers** and tuned the Linux TCP/IP networking stack for deterministic high-throughput, low-latency edge packet transmission across distributed sensor nodes.
- Configured and deployed **ChirpStack IoT server stacks** and LoRaWAN packet forwarders for enterprise industrial telemetry and edge gateway fleet monitoring.
- Collaborated with engineering teams to automate board bring-up scripts and flashing workflows, cutting multi-board regression testing cycles by 40%.

Research Intern (Computer Vision & Edge AI) – Indian Institute of Technology (IIT), Roorkee

Jun. 2025 – Jul. 2025

Roorkee, Uttarakhand, India

On-site

- Researched and developed an optimized real-time **Object Detection pipeline** based on custom-trained **YOLOv8** architectures for sustained low-latency edge video analytics.
- Accelerated deep learning inference on **Intel Core Ultra Neural Processing Units (NPUs)** leveraging the **Intel OpenVINO toolkit**, converting models to INT8/FP16 Intermediate Representations (IR).
- Benchmarked multi-device throughput and latency curves across CPU, integrated GPU, and dedicated NPU execution units, achieving continuous 60+ FPS inference with minimal power draw and thermal footprint.
- Built internal data annotation, dataset augmentation, and automated validation pipelines for custom computer vision models with OpenCV.

Technical Organizer & Systems Lead – Innovathon & University Research

Aug. 2024 – Present

University of Jammu, India

Leadership / Research

- Core technical organizer for **Innovathon 1.0 National Hackathon**, leading platform operations, server infrastructure, and mentoring 200+ collegiate participants on edge computing and machine learning.
- Architected telemetry ingestion pipelines and open-source scientific computing templates adopted across departmental research groups.
- Selected for dramatic portrayal as Arjuna and Guru Kripacharya in *Rashmirathi* at the **25th Bharat Rang Mahotsav (BRM)** International Theatre Festival (2026) organized by the National School of Drama (NSD) & Ministry of Culture.

PROJECTS

NeuroView — 3D Spatial Telemetry & WebGL <i>Python, FastAPI, Three.js, WebGL/GLSL, MNE, Groq, Docker</i>  	2026 – Present
- Engineered an asynchronous 3D spatial telemetry and neural signal visualization platform streaming multi-channel scientific data and 3D coordinate matrices. - Built an interactive Three.js WebGL rendering engine with custom GLSL shaders executing real-time Inverse Distance Weighting (IDW) scalar field heatmap interpolation across arbitrary 3D geometric surfaces. - Designed a high-throughput FastAPI backend utilizing NumPy and MNE-Python for parsing binary data streams, baseline noise filtering, and spatial electrode coordinate projection. - Integrated an automated diagnostic pipeline utilizing cascading agentic LLMs (Groq API, DeepSeek) for real-time anomaly detection and natural language telemetry reporting. - Delivered a fully containerized, cloud-native application using Docker with CI/CD optimization for microservice-based edge and cloud deployments.	
SlipSense — Res U-Net Landslide Mapping <i>PyTorch, DeepLabV3+, CRUNet, HDF5, QGIS, Flask</i>  	2025 – Present
- Developed an AI-powered semantic segmentation system to detect and delineate landslide hazards across complex Himalayan topography using 14-band multispectral satellite imagery (Sentinel-2 HDF5). - Designed a custom Residual U-Net (CRUNet) architecture integrated with Atrous Spatial Pyramid Pooling (ASPP) from DeepLabV3+ to capture multi-scale spatial receptive fields and eliminate vanishing gradients. - Implemented an end-to-end geospatial preprocessing pipeline in Python and QGIS for radiometric calibration, atmospheric correction, band stacking, and spatial alignment. - Exposed trained deep learning models via a high-throughput Flask REST API, enabling sub-second inference and automated hazard boundary mapping for disaster mitigation.	
YOLO-Based Live Object Detection on NPU <i>Python, YOLOv8, OpenVINO, NPU, OpenCV</i>  	2025
- Engineered a custom YOLOv8-based real-time object detection architecture optimized using Intel's OpenVINO toolkit for edge AI PC hardware. - Quantized model weights to INT8/FP16 Intermediate Representation (IR), deploying directly to Intel Core Ultra Neural Processing Units (NPUs) for ultra-low power consumption. - Implemented asynchronous frame processing with OpenCV, sustaining 60+ FPS high-throughput video streams without taxing CPU/GPU resources.	
Custom Raspberry Pi Router & Hardening <i>OpenWrt, Linux Kernel, iptables, Bash, DoH</i>  	2024
- Compiled and deployed custom OpenWrt Linux firmware on Raspberry Pi 3 hardware, creating a hardened network gateway. - Engineered robust iptables firewall policies, NAT masquerading, stateful packet inspection, and DNS-over-HTTPS (DoH).	
Temperature Prediction Platform (ETS) <i>Python, C (ANSI), Time Series ETS, Derivatives</i> 	2024 – 2025
Hybrid multi-language platform predicting daily average temperatures using Exponential Smoothing (ETS) in Python and discrete derivative calculation routines in ANSI C for rapid numerical analysis.	

CERTIFICATIONS & HONORS

AMD — AI Agents 101: Building AI Agents with MCP & Open-Source Inference	Issued: Aug. 2026
Skills: Model Context Protocol (MCP), Agentic Workflows, Open-Source Inference, Tool Calling, System Architecture	
UniConverge Technologies — Embedded & IoT Engineering Industrial Certification	Issued: Jul. 2026
Skills: Embedded Linux, OpenWrt, LoRaWAN, ChirpStack, Kernel Drivers, SPI Drivers, Hardware Prototyping	
IIT Kanpur — Junior Scientist Honor (26th National Children Science Congress)	Issued: Dec. 2018
Skills: Scientific Research Methodology, Experimental Design, State Representative, Technical Presentation	
IIT Jammu — Visual Attention Modeling & High-Speed Instrumentation Workshop	Issued: Dec. 2024
Skills: High-Speed Eye-Tracking, Visual Attention Modeling, Spatial Gaze Calibration, Statistical Research	
Ministry of Education & AICTE — IDE Innovation & Entrepreneurship Bootcamp	Issued: Jul. 2026
Skills: Deep-Tech Venture Design, Rapid Prototyping, Product Architecture, Technology Commercialization	